

AKILEN

1. NAME OF THE MEDICINAL PRODUCT

Akilen 40 mg film-coated tablets

Akilen 80 mg film-coated tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Film-coated tablets containing either 40mg or 80mg of verapamil hydrochloride.

Excipients with known effect:

Lactose (only for Akilen 40mg film-coated tablets). Each 40mg film-coated tablet contains 40mg lactose monohydrate.

Sunset yellow (only for Akilen 80mg film-coated tablets). Each 80mg film-coated tablet contains 0.197mg sunset yellow

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Film-coated tablet.

Akilen 40mg film-coated tablets: very pale green, round, convex film-coated tablets, with diameter of core 7.0mm.

Akilen 80mg film-coated tablets: orange, round, biconvex, with diameter of core 7.0mm.

4. CLINICAL PARTICULARS

4.1. Therapeutic indications

Prophylaxis and/or treatment of

- Angina pectoris, including Prinzmetal's angina (coronary spasm, vasospastic angina).
- Supraventricular tachycardias such as paroxysmal supraventricular tachycardia, atrial fibrillation/flutter with rapid ventricular response (except in [Wolff-Parkinson-White syndrome](#), see section 4.4)
- Mild to moderate essential hypertension.

4.2. Posology and method of administration

Angina:

Adults: 120mg is recommended, 80mg can be completely satisfactory in some patients with angina of effort. Less than 120mg is not likely to be effective in angina at rest and variant angina.

Supraventricular tachycardias: 40 – 120mg according to the severity of the condition

Children up to 2 years: 40mg tablet 2 – 3 times a day

2 years and above: 1 – 3 40mg tablets 2 – 3 times a day, according to age and effectiveness

Hypertension:

Adults: the usual dose range is 80 – 160mg. The dose should be increased from 80mg at weekly intervals according to response, either alone or in conjunction with another antihypertensive therapy. A further reduction in blood pressure may be obtained by combining verapamil hydrochloride with other antihypertensive agents e.g. thiazide diuretics

Children: up to 10mg/kg/day in divided doses according to severity of disease.

Method of administration

Tablets are for oral administration. They should be swallowed whole with a little water if necessary.

Verapamil should not be taken with grapefruit juice (see section 4.5).

4.3. Contraindications

- Hypersensitivity to the active substance or to any of the excipients listed in section 6.1
- Hypotension (of less than 90mmHg systolic)
- Second or third degree atrioventricular block; sick sinus syndrome (except in patients with a functioning artificial pacemaker); uncompensated heart failure; marked bradycardia (less than 50 beats/minute).
- Combination with beta-blockers is contraindicated in patients with poor ventricular function.
- Wolff-Parkinson-White syndrome.
- Concomitant ingestion of grapefruit juice is contraindicated.

- Acute myocardial infarction complicated by bradycardia, marked hypotension or left ventricular failure.
- Combination with ivabradine (see section 4.5)

4.4. Special warnings and precautions for use

Verapamil may affect left ventricular contractility as a result of its mode of action. The effect is small and not normally important, however, cardiac failure may be aggravated or precipitated if it exists. In cases with poor ventricular function, verapamil should therefore only be administered after appropriate therapy for cardiac failure such as digitalis, etc.

Verapamil may affect impulse conduction and should be administered with caution in patients with first degree atrioventricular block. The effects of verapamil and beta-blockers or other drugs may be additive both in respect of conduction and contraction, therefore care should be exercised when these are administered concurrently or closely together. This is especially true when either drug is administered intravenously.

Caution should be observed in the acute stage of myocardial infarction.

Patients with atrial fibrillation/flutter and an accessory pathway (*eg* Wolff-Parkinson-White syndrome) may rarely develop increased conduction across the anomalous pathway and ventricular tachycardia may be precipitated. Since verapamil is extensively metabolised in the liver, careful dose titration of verapamil is required in patients with liver disease. The disposition of verapamil in patients with renal impairment has not been fully established and therefore careful patient monitoring is recommended. Verapamil is not removed during dialysis.

AKILEN 40mg film-coated tablets contain lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine. AKILEN 80mg film-coated tablets contain sunset yellow, which may cause allergic reactions.

4.5. Interactions with other medicinal products and other forms of interaction

In vitro metabolic studies indicate that verapamil hydrochloride is metabolized by cytochrome P450 CYP3A4, CYP1A2, CYP2C8, CYP2C9 and CYP2C18. Verapamil has been shown to be an inhibitor of CYP3A4 enzymes and P-glycoprotein (P-gp). Clinically significant interactions have been reported with inhibitors of CYP3A4 causing elevation of plasma levels of verapamil hydrochloride while inducers of CYP3A4 have caused a lowering of plasma levels of verapamil hydrochloride, therefore, patients should be monitored for drug interactions.

H₂ Receptor antagonists

Cimetidine increases serum levels of verapamil.

Anticonvulsants:

Increased serum levels of carbamazepine which could lead to increased side-effects.
Reduced serum levels of verapamil when taken with phenytoin.

Barbiturates:

Reduced serum levels of verapamil when taken with Phenobarbital.

Lithium:

Serum levels of lithium may be reduced (pharmacokinetic effect). There may be increased sensitivity to lithium causing enhanced neurotoxicity (pharmacodynamic effect).

Alcohol:

Verapamil has been shown to increase the blood levels of alcohol and slow its elimination. Therefore, the effects of alcohol may be exaggerated.

Antihypertensive drugs:

Verapamil may have an additive effect with other antihypertensive drugs; thus, in many cases with verapamil, a reduction in the dosage of the other antihypertensive drug may be possible.

Beta blockers:

Verapamil may increase the plasma concentrations of metoprolol and propranolol which may lead to additive cardiovascular effects (e.g. AV block, bradycardia, hypotension, heart failure).

Intravenous beta-blockers should not be given to patients under treatment with verapamil.

Alpha blockers:

Verapamil may increase the plasma concentrations of prazosin and terazosin which may have an additive hypotensive effect.

Antiarrhythmics:

Verapamil may increase the plasma concentrations of quinidine. Pulmonary oedema may occur in patients with hypertrophic cardiomyopathy.

The combination of verapamil and antiarrhythmic agents may lead to additive cardiovascular effects (e.g. AV block, bradycardia, hypotension, heart failure).

Benzodiazepines and anxiolytics:

Verapamil may increase the plasma concentrations of midazolam.

Lipid lowering agents:

Verapamil may increase the plasma concentrations of atorvastatin, lovastatin and simvastatin.

Treatment with HMG CoA reductase inhibitors (e.g., simvastatin, atorvastatin or lovastatin) in a patient taking verapamil should be started at the lowest possible dose and titrated upwards. If verapamil treatment is to be added to patients already taking an HMG CoA reductase inhibitor (e.g., simvastatin, atorvastatin or lovastatin), consider a reduction in the statin dose and retitrate against serum cholesterol concentrations.

Atorvastatin has been shown to increase verapamil levels. Although there is no direct in vivo clinical evidence, there is strong potential for verapamil to significantly affect atorvastatin pharmacokinetics in a similar manner to simvastatin or lovastatin. Consider using caution when atorvastatin and verapamil are concomitantly administered.

Fluvastatin, pravastatin and rosuvastatin are not metabolized by CYP3A4 and are less likely to interact with verapamil.

Acetylsalicylic acid

Concomitant use of verapamil with aspirin may increase the risk of bleeding.

Dabigatran etexilate

When oral verapamil was co-administered with dabigatran etexilate (150mg), a P-gp substrate, the C_{max} and AUC of dabigatran were increased but magnitude of this change differs depending on time between administration and the formulation of verapamil. When verapamil 120mg immediate-release was co-administered one hour before a single dose of dabigatran etexilate, the dabigatran C_{max} was increased by about 180% and AUC by about 150%. No meaningful interaction was observed when verapamil was administered 2 hours after dabigatran etexilate (increase of C_{max} by about 10% and AUC by about 20%).

Close clinical surveillance is recommended when verapamil is combined with dabigatran etexilate and particularly in the occurrence of bleeding, notably in patients having a mild to moderate renal impairment.

Anti-infectives

Reduced serum levels of verapamil when taken with rifampicin.

Erythromycin, clarithromycin and telithromycin may increase the plasma concentrations of verapamil.

Anaesthesia:

Long-term verapamil therapy may give rise to potentiation of neuromuscular blocking agents during anaesthesia. Interactions have been reported during the concomitant use of verapamil and the following medications:

Theophylline:

Increased serum levels of theophylline which could lead to increased side-effects.

Immunosuppressants

Verapamil may increase the plasma concentrations of ciclosporin, everolimus, sirolimus and tacrolimus which could lead to increased side-effects.

Digoxin:

Verapamil has been shown to increase the serum concentration of digoxin and caution should therefore be exercised with regard to digitalis toxicity.

Dantrolene

Concomitant use of verapamil with intravenous dantrolene may cause hypotension, myocardial depression, and hyperkalaemia. This combination should be avoided.

Colchicine

Colchicine is a substrate for both CYP3A and the efflux transporter, P-glycoprotein (P-gp). Verapamil is known to inhibit CYP3A and P-gp. When verapamil and colchicine are administered together, inhibition of P-gp and/or CYP3A by verapamil may lead to increased exposure to colchicine. Combined use is not recommended.

Other:

An increase in verapamil serum level has been reported when taken with grapefruit juice.

Concomitant use with ivabradine is contraindicated due to the additional heart rate lowering effect of verapamil to ivabradine (see section 4.3).

4.6. Fertility, pregnancy and lactation

Animal studies have shown no teratogenic effect. Verapamil should not be administered during pregnancy unless, in the clinician's judgement, it is essential for the welfare of the patient. The possibility that verapamil can cause relaxation of the uterine muscle should be considered at term.

Verapamil is excreted in breast milk at concentrations approximately equal to 0.5-1.0 times that co-existing in maternal plasma. However, the amount ingested in such circumstances is less than 1% of the recommended therapeutic infant dose, assuming normal suckling rates.

4.7. Effects on ability to drive and use machines

Depending on individual susceptibility, the patient's ability to drive or operate machines may be impaired due to feelings of drowsiness. This is particularly true in the initial stages of treatment, or when changing over from another drug. Verapamil has been shown to increase the blood levels of alcohol and slow its elimination. Therefore, the effects of alcohol may be exaggerated.

4.8. Undesirable effects

Immune system disorders: allergic reactions (e.g. erythema, pruritus, urticaria) are very rarely seen.

Nervous system disorders: headaches occur rarely, dizziness, paraesthesia, tremor, extrapyramidal syndrome (e.g. parkinsonism), dystonia.

Ear and labyrinth disorders: vertigo, tinnitus.

Cardiac disorders: bradycardic arrhythmias such as sinus bradycardia, sinus arrest with asystole, 2nd and 3rd degree AV block, bradyarrhythmia in atrial fibrillation, palpitations, tachycardia, development or aggravation of heart failure, hypotension.

Vascular disorders: flushing, peripheral oedema.

Gastrointestinal disorders: nausea, vomiting, constipation is not uncommon, ileus and abdominal pain/discomfort. Gingival hyperplasia may very rarely occur when the drug is administered over prolonged periods. This is fully reversible when the drug is discontinued.

Skin and subcutaneous tissue disorders: alopecia, ankle oedema, Quincke's oedema, Steven-Johnson syndrome, erythema multiforme, erythromelalgia, purpura.

Musculoskeletal and connective tissue disorders: muscular weakness, myalgia and arthralgia.

Reproductive system and breast disorders: impotence (erectile dysfunction) has been rarely reported and isolated cases of galactorrhoea. Gynaecomastia was observed on very rare occasions in elderly male patients under longer term verapamil treatment which was fully reversible in all cases when the drug was discontinued.

General disorders and administration site conditions: fatigue.

Investigations: On very rare occasions, a reversible impairment of liver function characterised by an increase in transaminases and/or alkaline phosphatase, may occur during verapamil treatment and is most probably a hypersensitivity reaction.

4.9. Overdose

The course of symptoms in verapamil intoxication depends on the amount taken, the point in time at which detoxification measures are taken and myocardial contractility (age-related). The main symptoms are as follows: blood pressure fall (at times to values not detectable), shock symptoms, loss of consciousness, 1st and 2nd degree AV block (frequently as Wenckebach's phenomenon with or without escape rhythms), total AV block with total AV dissociation, escape rhythm, asystole, bradycardia up to high degree AV block and, sinus arrest, hyperglycaemia, stupor and metabolic acidosis. Fatalities have occurred as a result of overdose.

The therapeutic measures to be taken depend on the point in time at which verapamil was taken and the type and severity of intoxication symptoms. In intoxications with large amounts of slow-release preparations, it should be noted that the release of the active drug and the absorption in the intestine may take more than 48 hours. Verapamil hydrochloride cannot be removed by haemodialysis. Depending on the time of ingestion, it should be taken into account that there may be some lumps of incompletely dissolved tablets along the entire length of the gastrointestinal tract, which function as active drug depots.

General measures to be taken: Gastric lavage with the usual precautions, even later than 12 hours after ingestion, if no gastrointestinal motility (peristaltic sounds) is detectable. Where intoxication by a modified release preparation is suspected, extensive elimination measures are indicated, such as induced vomiting, removal of the contents of the stomach and the small intestine under endoscopy, intestinal lavage, laxative, high enemas. The usual intensive resuscitation measures apply, such as extrathoracic heart massage, respiration, defibrillation and/or pacemaker therapy.

Specific measures to be taken: Elimination of cardiodepressive effects, hypotension or bradycardia. The specific antidote is calcium, e.g. 10-20ml of a 10% calcium gluconate solution administered intravenously (2.25 - 4.5mmol), repeated if necessary or given as a continuous drip infusion (e.g. 5mmol/hour).

The following measures may also be necessary: In case of 2nd or 3rd degree AV block, sinus bradycardia, asystole - atropine, isoprenaline, orciprenaline or pacemaker therapy. In case of hypotension - dopamine, dobutamine, noradrenaline (norepinephrine). If there are signs of continuing myocardial failure - dopamine, dobutamine, if necessary repeated calcium injections.

5. PHARMACOLOGICAL PROPERTIES

5.1. Pharmacodynamic properties

Pharmacotherapeutic group: Selective calcium channel blockers with direct cardiac effects, phenylalkylamine derivatives. ATC code: C08 DA01

Verapamil hydrochloride is a calcium channel blocker and is classified as a class IV anti-arrhythmic agent.

Mechanism of action

Verapamil inhibits the entry of calcium into smooth muscle cells of the systemic and coronary arteries and in the cells of cardiac muscle and the intracardiac conduction system.

Verapamil lowers peripheral vascular resistance with little or no reflex tachycardia. Its efficacy in reducing both raised systolic and diastolic blood pressure is thought to be primarily due to this mode of action.

The decrease in systemic and coronary vascular resistance and the sparing effect on intracellular oxygen consumption appear to explain the anti-anginal properties of the product.

Due to the effect on the movement of calcium in the intracardiac conduction system, verapamil reduces automaticity, decreases conduction velocity and increases the refractory period.

5.2. Pharmacokinetic properties

Absorption

Verapamil is approximately 90% absorbed from the gastrointestinal tract.

Distribution

Verapamil acts within 1-2 hours after oral administration with a peak plasma concentration after 1-2 hours. There is considerable interindividual variation in plasma concentrations. Verapamil is about 90% bound to plasma proteins.

Biotransformation

Verapamil is subject to very considerable first-pass metabolism in the liver and the bioavailability is only about 20%. It is extensively metabolised in the liver to at least 12 metabolites of which norverapamil has been shown to have some activity.

Elimination

Verapamil exhibits bi- or tri-phasic elimination kinetics and is reported to have a terminal plasma half-life of 2-8 hours following a single oral dose. After repeated oral doses this increases to 4.5-12 hours. About 70% of a dose

is excreted by the kidneys in the form of its metabolites but about 16% is also excreted in the bile into the faeces. Less than 4% is excreted unchanged.

Pregnancy and breast-feeding

Verapamil crosses the placenta and is excreted in breast milk (see section 4.6).

6. PHARMACEUTICAL PARTICULARS

6.1. List of excipients

Akilen 40mg film-coated tablets:

- Cellulose microcrystalline
- Lactose monohydrate
- Croscarmellose sodium
- Silica Colloidal Anhydrous
- Magnesium stearate
- Opadry green OY-3875H (hypromellose (E464), Titanium dioxide (E171), Macrogol/polyethylene glycol 400, Iron oxide yellow (E172), Indigo carmine aluminium lake (E132), Iron oxide black (E172), PEG 6000

Akilen 80 mg film-coated tablets:

- Cellulose microcrystalline
- croscarmellose sodium,
- silica colloidal anhydrous,
- magnesium stearate,
- Opadry orange OY – 2500 (hypromellose (E464), titanium dioxide (E171), macrogol/Polyethylene glycol 400, sunset yellow lake (E110)), PEG 6000

6.2. Incompatibilities

None known.

6.3. Shelf life

5 years

6.4. Special precautions for storage

Store below 30°C, in the original package

6.5. Nature and contents of container

Combination polyvinylchloride film and aluminium foil blisters of ten tablets.

Akilen 40mg film-coated tablets: Blisters are placed, with a patient information leaflet, in a card carton. Packs of 100, 500 and 1000 tablets are available.

Akilen 80mg film-coated tablets: Blisters are placed, with a patient information leaflet, in a card carton. Packs of 500 tablets are available.

Not all pack sizes may be marketed.

7. Product Registration Holder:

Komedic Sdn Bhd, 4 Jalan PJS 11/14, Bandar Sunway, 46150 Petaling Jaya

8. Manufacturer:

Medochemie Ltd., 1-10 Constantinoupoleos Street, 3011 Limassol, Cyprus

9. DATE OF REVISION OF THE TEXT

06/2021